HARE: History-Assisted Residual Estimation of Long-Term Treatment Effects with Limited Target Outcomes

Siyu Chen
Renmin University of China

Abstract

We study long-term average treatment effect estimation when short-term measures are widely observed, target outcomes are partially observed, and historical outcome data are available. History-Assisted Residual Estimation (HARE) combines a historical outcome predictor with treatment-specific residual corrections learned from target labels within a cross-fitted, surrogate-assisted estimator. Under unconfoundedness, missing at random, and positivity, a reference estimator removes historical-index bias with known treatment and labeling probabilities, without requiring statistical surrogacy or cross-population comparability. In a finite-dimensional least-squares benchmark with a known historical function, a correctly specified lower-dimensional residual model reduces outcome-regression error and target-label requirements. Monte Carlo simulations evaluate HARE-C, a constant-correction implementation. Under constant discrepancies at an expected target-label budget of 800, empirical bias is 0.0103 for HARE-C, compared with -0.5201 for the surrogate-index benchmark. Its root mean squared error is 40.8% lower than the target-only comparator's at the same budget and remains below that comparator's at twice the budget. At the 800-label budget, the relative advantage narrows as unmodeled heterogeneity increases. These findings highlight the complementary roles of historical prediction and limited target outcomes in improving long-term effect estimation.

1. Introduction and Related Work

1.1 Limited Outcome Observations and Historical Information

Evaluating the long-term effects of policies or interventions is constrained by the costs of outcome measurement and data acquisition. Follow-up surveys, administrative record linkage, and other verification procedures may cover only a subset of study participants, whereas more readily available short-term measures are observed for most participants. Researchers must therefore combine limited observations of long-term outcomes with richer short-term information to estimate the long-term average treatment effect in the target population (Kallus and Mao, 2025).

Consider an illustrative evaluation of a job-training program. Researchers seek to estimate the program's effect on participants' earnings three years after treatment. Treatment assignments, pretreatment characteristics, and employment and earnings records from the first post-treatment year are available for all participants. The three-year follow-up period has ended, but survey budgets and record-linkage capacity permit verification of three-year earnings for only some participants. Meanwhile, programs in other locations or earlier training initiatives have already collected data on individual characteristics, early employment outcomes, and long-term

earnings. The available information then consists of three components: historical data from programs in other locations or earlier cohorts with observed short- and long-term outcomes, a labeled subset of the target sample, and the remaining target observations with only short-term measures.

Historical data contain rich predictive relationships between short- and long-term outcomes. However, differences in program delivery and local labor-market conditions may change how early employment and earnings translate into long-term earnings in the target program. Observed target outcomes provide direct information about the population of interest, but are available for only a subset of its members. The challenge is to combine these limited target outcomes with historical information to estimate the target program's long-term effect accurately despite differences in outcome relationships.

1.2 Two Surrogate-Based Estimation Approaches

Relative to the setting in which both historical information and target long-term outcomes are available, existing research has studied two endpoint information structures. In the first, target long-term outcomes are entirely unobserved, but external long-term outcome data are available. In the second, target long-term outcomes are partially observed, and estimation relies on information within the target sample without an additional historical outcome sample. The surrogate-index approach and surrogate-assisted estimation with partially observed outcomes provide important frameworks for these respective settings (Athey et al., 2026; Kallus and Mao, 2025).

The first approach is the surrogate-index method. Athey et al. (2026) consider two samples: an experimental sample containing treatment assignments, short-term outcomes, and pretreatment characteristics, but no long-term outcomes; and an external observational sample containing short-term outcomes, pretreatment characteristics, and long-term outcomes, but no treatment assignment information. The method first learns the conditional expectation of the long-term outcome given short-term outcomes and pretreatment characteristics in the external sample. This prediction forms a surrogate index. The average treatment effect on that index is then estimated in the experimental sample. Under the relevant assumptions, the treatment effect on the index equals the treatment effect on the actual long-term outcome.

However, this equivalence relies on strong assumptions. In addition to unconfounded treatment assignment in the experimental sample and the requisite support conditions, *statistical surrogacy* requires the long-term outcome and treatment status to be conditionally independent given short-term outcomes and pretreatment characteristics. *Comparability* requires the experimental and historical populations to have the same conditional distribution of the long-term outcome given these variables.

Let Y denote the long-term outcome, S the short-term measures, X pretreatment characteristics, and T treatment status. Let $P \in \{E, H\}$ indicate whether an observation comes from the target experimental or historical population. The two assumptions are

$$\begin{aligned} \text{Statistical surrogacy:} \quad & Y \perp T \mid X, S, P = E, \\ \text{Comparability:} \quad & Y \perp P \mid X, S. \end{aligned}$$

Strong predictive power of the short-term measures does not by itself establish surrogacy, nor do common measurement definitions establish comparability. Athey et al. (2026) discuss the strength of these assumptions and characterize the biases and bounds arising when they fail.

When surrogacy or comparability fails, the surrogate-index estimator can still target a meaningful causal effect under unconfounded treatment assignment: the effect of treatment on the historical prediction index.

This effect, however, need not equal the long-term outcome effect of interest. When the two differ, the estimator is systematically biased for the long-term effect. The bias reflects a difference between the estimand and the causal target and persists even with arbitrarily large samples and an exactly learned historical outcome regression (Athey et al., 2026).

The second approach directly uses partially observed target long-term outcomes. Kallus and Mao (2025) study a setting in which treatment assignments, pretreatment characteristics, and short-term outcomes are observed for all units, while the long-term outcome is observed only for a subset. Rather than requiring short-term outcomes to replace the long-term outcome completely, they treat them as auxiliary information that can improve precision. Under unconfounded treatment assignment, missing-at-random outcomes, and the relevant support conditions, their framework allows treated and control units with the same short-term outcomes and pretreatment characteristics to have different long-term outcome relationships. They construct a cross-fitted estimator and establish robustness, asymptotic normality, and efficiency results under appropriate convergence conditions.

This framework already provides a systematic approach to scarce long-term outcomes, but the limited target labels must still support learning of the outcome regressions. When pretreatment covariates and short-term outcomes are rich and their relationships with the primary outcome are complex, this learning task may be subject to substantial finite-sample error. Kallus and Mao (2025) emphasize that when the unlabeled sample is much larger than the labeled sample, the number of labeled observations remains an important bottleneck for accurate ATE estimation. When external long-term outcome information is already available, a natural further question is how that information can reduce the learning burden placed on limited target labels.

1.3 Research Question and History-Assisted Residual Estimation

This paper asks: *When historical outcome information and a limited number of target long-term outcomes are both available, how can they be combined to improve estimation of the target population's long-term average treatment effect and reduce the target-label requirement for a given level of accuracy?*

We consider an information structure comprising a historical sample, a labeled target subsample, and an unlabeled target subsample. Relative to a purely historical surrogate approach, this structure supplies actual target outcomes with which to correct the effect omitted by historical prediction. Relative to a partially labeled approach using only target data, it supplies additional historical outcome information to assist outcome-regression learning. The two sources play complementary roles: historical data provide a rich predictive foundation, while target labels provide direct information about the outcome relationship in the target population.

Let $m_t^E(x, s)$ denote the expected long-term outcome for target units with treatment status $T = t$, pretreatment characteristics $X = x$, and short-term measures $S = s$. We write

$$\begin{aligned} m_t^E(x, s) &= \mathbb{E}_E[Y \mid T = t, X = x, S = s] \\ &= h_H(x, s) + d_t(x, s), \end{aligned} \tag{1.1}$$

where $h_H(x, s) = \mathbb{E}_H[Y \mid X = x, S = s]$ is the historical conditional mean and $d_t(x, s)$ is the treatment-specific discrepancy between the target and historical conditional means. We focus on settings in which historical data capture complex predictive structure shared with the target population, while the remaining target-specific discrepancy can be approximated by a simpler correction model. Target labels can then be used to learn this correction rather than the entire outcome regression.

On this basis, we study *History-Assisted Residual Estimation* (HARE). The method first trains a baseline predictor using historical data, then uses observed long-term outcomes in the target training sample to learn a residual correction. The combined prediction enters a cross-fitted estimating equation based on the framework of Kallus and Mao (2025). The final estimator retains the weighted residual correction using actual target outcomes, rather than treating historical predictions as observed long-term outcomes. Our numerical implementation fits a constant residual model separately within the labeled training observations of each treatment arm; we denote this implementation by HARE-C.

1.4 Related Literature, Contributions, and Main Findings

This paper also relates to transfer learning and prediction-assisted causal inference. In high-dimensional linear regression, Li et al. (2022) use sparsity of the contrast between source and target regression models to characterize the informativeness of auxiliary data, and establish improvements in target prediction and estimation under appropriate conditions. Demirel et al. (2024) study how predictions trained on external observational data can assist causal generalization from a limited randomized trial, using trial data to learn a bias correction for the external predictions. Following the idea that external prediction and target residual information can complement one another, we focus on long-term ATE estimation when post-treatment short-term measures are widely observed but target long-term outcomes are only partially observed.

Our work makes three contributions. First, we develop HARE for long-term treatment-effect estimation with historical outcome data and partially observed target outcomes, and implement it using treatment-specific constant residual corrections.

Second, we establish two complementary analytical results. We construct a target-outcome correction that eliminates historical-index bias when treatment and labeling probabilities are known. In a finite-dimensional least-squares benchmark with a known historical prediction, we also show that learning a lower-dimensional residual improves outcome-regression accuracy and reduces the number of target labels required to attain a given prediction error.

Third, Monte Carlo simulations evaluate the resulting gains in long-term ATE estimation. We compare HARE-C with historical-index and target-only benchmarks, assessing bias reduction, RMSE at a common target-label budget, and the target-label budgets required to attain comparable ATE accuracy. We further examine how these gains vary with target-label availability and the magnitude of heterogeneous historical–target discrepancies.

The simulations illustrate the complementary value of historical information and target labels. In the main experiment, the historical and target samples each contain 8,000 observations, the true ATE is 2, and historical–target discrepancies are treatment-specific constants. Each configuration is repeated 50 times. At an expected target-label budget of 800, the empirical bias of the surrogate-index benchmark is -0.5201, compared with 0.0103 for HARE-C. The root mean squared errors of the target-only GBM implementation and HARE-C are 0.2392 and 0.1416, respectively. In the same constant-discrepancy setting, the target-only method has an RMSE of 0.1549 at a label budget of 1,600, still above that of HARE-C at a budget of 800. Thus, conditional on the availability of historical data, HARE-C attains an empirical RMSE no greater than that of the comparator using half the expected target-label budget in this configuration.

Additional experiments characterize the method's range of applicability. We focus on settings in which historical data contain predictive structure that can be reused and target long-term labels are relatively scarce. As the number of target labels increases, the target-only method learns the outcome relationship more fully, and the relative advantage of historical assistance narrows in the larger-label experiments. When historical–target

heterogeneity that cannot be represented by a constant correction becomes stronger, both HARE-C's prediction error and its treatment-effect estimation error increase, and its relative accuracy advantage diminishes. These findings indicate that the value of historical information depends on the scarcity of target labels and on how effectively the chosen correction model captures the target discrepancy.

2. Data Structure, Estimand, and Modeling Conditions

2.1 Data Sources and Estimand

Let E denote the target population and H the historical or external population. Expectations under these populations are denoted by $\mathbb{E}_E[\cdot]$ and $\mathbb{E}_H[\cdot]$, respectively, and probabilities under the target population by $\mathbb{P}_E(\cdot)$.

For an individual unit, let X be a vector of pretreatment characteristics and $T \in \{0, 1\}$ the treatment indicator, with $T = 1$ denoting treatment and $T = 0$ control. Let S be a vector of observed post-treatment short-term measures and Y the long-term outcome of interest. Lowercase x, t, and s denote realizations of the corresponding variables. In the target sample, let $R \in \{0, 1\}$ indicate whether the long-term outcome is observed: Y is observed when $R = 1$ and missing when $R = 0$, while X, T, S remain observed. This target-sample observation structure follows the partially observed outcome framework of Kallus and Mao (2025).

The historical sample contains X, S, Y and does not require treatment status to be available. The target sample is partitioned by R into a labeled subsample E_L and an unlabeled subsample E_U. Let N_H and N_E denote the historical and target sample sizes, and let N_L and N_U denote the numbers of labeled and unlabeled target observations, so that $N_E = N_L + N_U$.

Table 1. Observed information by sample component

Sample component	Observed variables	Sample size
Historical sample H	X, S, Y	N_H
Labeled target subsample E_L	X, T, S, Y; $R = 1$	N_L
Unlabeled target subsample E_U	X, T, S; $R = 0$	N_U

In the theoretical sampling setup, the historical and target samples are independent, with each consisting of independent and identically distributed observations from its respective population. The sets E_L and E_U are components of the same target sample induced by the outcome-observation mechanism, not two additional independently sampled populations. Their covariate distributions may differ. Variables shared across the historical and target samples have consistent definitions, measurement conventions, and outcome horizons, but the populations need not share the same distributions or outcome relationships. Consistent measurement of shared variables is also a basic requirement for combining surrogate information across samples (Athey et al., 2026).

For a unit in the target population, denote the potential long-term outcome under treatment t by $Y(t)$ and the corresponding potential short-term measures by $S(t)$. The estimand is the long-term average treatment effect in the target population:

$$\tau = \mathbb{E}_E[Y(1) - Y(0)]. \tag{2.1}$$

This estimand concerns the entire target population, not only the subgroup with observed long-term outcomes. Historical data provide auxiliary information without changing the population of interest.

2.2 Identification Conditions

We distinguish identification conditions for the target effect from structural conditions under which historical information can improve learning. The former follow the treatment-assignment and missing-data framework of Kallus and Mao (2025) and apply to the entire target population, including labeled and unlabeled units. All independence statements below refer to the target population; $\perp$ denotes conditional independence.

Assumption A1 (Consistency and no interference)**.** Treatment is well defined, and a unit's outcomes are unaffected by the treatment assignments of other units. Realized long-term outcomes and short-term measures equal the potential values corresponding to the realized treatment:

$$Y = Y(T), \qquad S = S(T). \tag{2.2}$$

Realization and observability are distinct: even if $R = 0$, $Y(T)$ is the unit's realized outcome, although it is not observed by the researcher.

Assumption A2 (Unconfounded treatment assignment)**.** Conditional on pretreatment characteristics X, treatment assignment is independent of the corresponding potential long-term outcomes and short-term measures:

$$\big(Y(t), S(t)\big) \perp T \mid X, \qquad t \in \{0, 1\}. \tag{2.3}$$

Treatment probabilities may depend on pretreatment characteristics; completely randomized assignment is not required (Kallus and Mao, 2025).

Assumption A3 (Long-term outcomes missing at random)**.** Conditional on treatment status, pretreatment characteristics, and the corresponding short-term measures, outcome observability is independent of the potential long-term outcome:

$$R \perp Y(t) \mid T, X, S(t), \qquad t \in \{0, 1\}. \tag{2.4}$$

Label acquisition may depend on observed information, but that information must account for the association between outcome observability and the long-term outcome. Labeled units with the same T, X, S can consequently inform the outcome relationship for unlabeled units (Kallus and Mao, 2025).

Assumption A4 (Positivity and moment conditions)**.** Let $e(x)$ be the treatment probability conditional on pretreatment characteristics, and let $r(t, x, s)$ be the labeling probability conditional on treatment, pretreatment characteristics, and short-term measures:

$$\begin{aligned} e(x) &= \mathbb{P}_E(T = 1 \mid X = x), \\ r(t, x, s) &= \mathbb{P}_E(R = 1 \mid T = t, X = x, S = s). \end{aligned} \tag{2.5}$$

For point identification, require on the relevant support of the target population that

$$0 < e(x) < 1, \qquad 0 < r(t, x, s) \leq 1. \tag{2.6}$$

Target potential outcomes and historical outcomes have finite second moments, so that the mean squared errors considered below are defined. Additional weighted-moment and convergence conditions required for

finite variance of inverse-probability-weighted terms and asymptotic inference are stated with the relevant results. Positivity differs from imposing uniform bounds away from zero, which is stronger. Kallus and Mao (2025) separately study regimes with a positive limiting labeled fraction and regimes in which that fraction vanishes.

These conditions constrain the true data-generating and observation mechanisms; they are not automatic guarantees about fitted models. Fitting logistic regressions for e or r, or clipping estimated probabilities, does not establish unconfoundedness, missing at random, or the required support conditions.

2.3 Historical Prediction and Target Discrepancy

Let $h_H(x, s)$ denote the conditional mean of the historical long-term outcome given pretreatment characteristics and short-term measures:

$$h_H(x, s) = \mathbb{E}_H[Y \mid X = x, S = s]. \tag{2.7}$$

This function describes the historical predictive relationship and corresponds to the external outcome conditional expectation used in the surrogate-index approach (Athey et al., 2026).

Let $m_t^E(x, s)$ denote the target outcome conditional mean that additionally conditions on treatment status:

$$m_t^E(x, s) = \mathbb{E}_E[Y \mid T = t, X = x, S = s]. \tag{2.8}$$

Under consistency, missing-at-random outcomes, and label positivity, this function is identified from the corresponding conditional mean among labeled target observations:

$$m_t^E(x, s) = \mathbb{E}_E[Y \mid T = t, X = x, S = s, R = 1]. \tag{2.9}$$

It corresponds to the outcome regression conditional on treatment, pretreatment characteristics, and surrogates in Kallus and Mao (2025).

Define the discrepancy between these outcome relationships as

$$d_t(x, s) = m_t^E(x, s) - h_H(x, s). \tag{2.10}$$

We therefore use the representation

$$m_t^E(x, s) = h_H(x, s) + d_t(x, s). \tag{2.11}$$

The historical function supplies the baseline prediction; the target discrepancy describes the adjustment required for the target treatment status and individual characteristics.

To average individual short-term information at the level of pretreatment characteristics, define $\mu_t^E(x)$ by integrating the target outcome regression over the target distribution of short-term measures:

$$\mu_t^E(x) = \mathbb{E}_E[m_t^E(x, S) \mid T = t, X = x]. \tag{2.12}$$

The averaging distribution is that of S under the relevant treatment status in the target population, not the historical distribution of S. Under the identification conditions above,

$$\mu_t^E(x) = \mathbb{E}_E[Y(t) \mid X = x], \tag{2.13}$$

and hence

$$\tau = \mathbb{E}_E[\mu_1^E(X) - \mu_0^E(X)]. \tag{2.14}$$

This nested conditional-expectation structure is consistent with the identification representation in Kallus and Mao (2025).

Modeling Relaxation (Allowing nonzero target discrepancies)**.** We relax the restriction $d_t(x, s) \equiv 0$, allowing the target outcome regression to differ from the historical conditional mean.

To relate this relaxation to the surrogate-index assumptions, define the target conditional mean that pools treatment groups:

$$\bar{m}^E(x, s) = \mathbb{E}_E[Y \mid X = x, S = s]. \tag{2.15}$$

The target discrepancy can then be decomposed as

$$\begin{aligned} d_t(x, s) = \quad & \underbrace{m_t^E(x, s) - \bar{m}^E(x, s)}_{\text{treatment-specific mean discrepancy}} \\ + \quad & \underbrace{\bar{m}^E(x, s) - h_H(x, s)}_{\text{cross-population mean discrepancy}} \quad . \end{aligned} \tag{2.16}$$

Statistical surrogacy and comparability impose distinct restrictions on these components (Athey et al., 2026):

$$\begin{aligned} \text{Statistical surrogacy} \quad &\Longrightarrow \quad m_t^E(x, s) - \bar{m}^E(x, s) = 0, \qquad t \in \{0, 1\}, \\ \text{Comparability} \quad &\Longrightarrow \quad \bar{m}^E(x, s) - h_H(x, s) = 0. \end{aligned}$$

When both assumptions hold,

$$m_t^E(x, s) = \bar{m}^E(x, s) = h_H(x, s) \quad \Longrightarrow \quad d_t(x, s) = 0, \qquad t \in \{0, 1\}.$$

Our framework allows either conditional-mean restriction to fail. The resulting discrepancy is captured by $d_t(x, s)$, and observed target long-term outcomes provide information for its estimation and correction.

2.4 Structural Conditions for Historical-Assisted Learning

For any square-integrable function $f(x, s)$, define its squared norm under the target distribution corresponding to treatment state t as

$$\|f\|_{E,t}^2 = \mathbb{E}_E[f(X, S(t))^2]. \tag{2.17}$$

The squared norm of the difference between two prediction functions is their mean squared difference under this distribution. We consider two structural conditions supporting historical-assisted learning.

Structural Condition H1 (Accurate learning of the historical function)**.** Let $\widehat{h}_H$ denote the predictor trained on the historical sample. Historical data cover the target-relevant regions of X, S and permit estimation of the historical function with sufficient accuracy that the learning errors

$$\left\| \widehat{h}_H - h_H \right\|_{E,t}^2, \qquad t \in \{0, 1\},$$

are small.

Structural Condition H2 (A simpler function class approximates the target discrepancy)**.** Let $\mathcal{F}$ be the function class considered for directly learning the complete target outcome regression, and let $\mathcal{G}$ be the correction class used to learn the target discrepancy. Relative to $\mathcal{F}$, $\mathcal{G}$ has lower learning complexity and can approximate d_t effectively. For an approximation-error tolerance $a_t \geq 0$, require

$$\inf_{g \in \mathcal{G}} \|d_t - g\|_{E,t}^2 \leq a_t^2. \tag{2.18}$$

The two conditions must be considered jointly. Because the historical prediction is estimated, the residual relationship facing target labels is

$$m_t^E(x,s) - \widehat{h}_H(x,s) = d_t(x,s) + \left[h_H(x,s) - \widehat{h}_H(x,s)\right]. \tag{2.19}$$

Target residual learning must therefore accommodate both the true historical–target discrepancy and the historical function's estimation error. When the historical prediction is sufficiently accurate and the target discrepancy admits a relatively simple approximation, limited target labels may avoid relearning the entire complex outcome relationship.

3. History-Assisted Residual Estimation

3.1 The Kallus–Mao Estimation Framework

Under the identification conditions in Section 2, the target long-term average treatment effect satisfies

$$\tau = \mathbb{E}_E[\mu_1^E(X) - \mu_0^E(X)]. \tag{3.1}$$

Building on this identification representation, Kallus and Mao (2025) construct a cross-fitted estimator using the efficient influence function. Partition the target sample into K folds, with I_k denoting evaluation fold k and I_{-k} the remaining observations. All function estimates indexed by k are trained on I_{-k}. In our notation, their estimator is

$$\begin{aligned}\widehat{\tau}_{\mathrm{KM}} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|I_k|}\sum_{i\in I_k}\Bigg[&\widehat{\mu}_{1,k}^E(X_i) - \widehat{\mu}_{0,k}^E(X_i)\\ &+ \frac{T_i - \widehat{e}_k(X_i)}{\widehat{e}_k(X_i)[1-\widehat{e}_k(X_i)]}\left\{\widehat{m}_{T_i,k}^E(X_i,S_i) - \widehat{\mu}_{T_i,k}^E(X_i)\right\}\\ &+ \frac{T_i R_i}{\widehat{e}_k(X_i)\widehat{r}_k(1,X_i,S_i)}\left\{Y_i - \widehat{m}_{1,k}^E(X_i,S_i)\right\}\\ &- \frac{(1-T_i)R_i}{[1-\widehat{e}_k(X_i)]\widehat{r}_k(0,X_i,S_i)}\left\{Y_i - \widehat{m}_{0,k}^E(X_i,S_i)\right\}\Bigg].\end{aligned} \tag{3.2}$$

Terms involving the long-term outcome are set to zero when $R_i = 0$. Equation (3.2) corresponds to Definition 2 in Kallus and Mao (2025) and comprises a conditional-outcome-mean contrast, a surrogate adjustment, and a long-term-outcome residual correction. We build on this structure by using historical-assisted predictions to estimate m_t^E and μ_t^E.

3.2 Historical-Assisted Outcome Learning and the Constant Implementation

First, train the historical predictor $\widehat{h}_H(x, s)$ on the independent historical sample. For each evaluation fold I_k, use labeled observations in the target training sample I_{-k} to fit treatment-specific regressions of $Y_i - \widehat{h}_H(X_i, S_i)$ on (X_i, S_i). Denote the fitted residual functions by $\widehat{d}_{t,k}(x, s)$. The historical-assisted outcome regression is

$$\widehat{m}^E_{t,k}(x, s) = \widehat{h}_H(x, s) + \widehat{d}_{t,k}(x, s). \tag{3.3}$$

HARE-C restricts each residual regression to a constant function:

$$\widehat{d}_{t,k}(x, s) \equiv \widehat{\Delta}_{tk}.$$

Let n_{tk} denote the number of training observations in I_{-k} with $T_i = t$ and $R_i = 1$. The fitted constant is

$$\widehat{\Delta}_{tk} = \frac{1}{n_{tk}} \sum_{\substack{i \in I_{-k} \\ T_i = t,\ R_i = 1}} \left\{ Y_i - \widehat{h}_H(X_i, S_i) \right\}. \tag{3.4}$$

This is an unweighted, intercept-only least-squares fit within each labeled treatment arm.

The second outcome regression averages the historical prediction over short-term measures, conditional on treatment and pretreatment characteristics, and adds the same fitted constant. Let $\widehat{\mathbb{E}}_{-k}[\cdot \mid T = t, X = x]$ denote a conditional-mean regression on X, fitted using all target training observations in I_{-k} with $T = t$. HARE-C therefore supplies the estimating equation with

$$\begin{aligned} \widehat{m}^E_{t,k}(x, s) &= \widehat{h}_H(x, s) + \widehat{\Delta}_{tk}, \\ \widehat{\mu}^E_{t,k}(x) &= \widehat{\mathbb{E}}_{-k}\left[\widehat{h}_H(X, S) \mid T = t, X = x \right] + \widehat{\Delta}_{tk}. \end{aligned} \tag{3.5}$$

The conditional-mean regression uses the historical predictions as its response and includes both labeled and unlabeled target observations. The historical predictor and this conditional-mean regression are fitted by gradient boosting; the residual correction is the constant in Equation (3.4).

3.3 Substitution into the Estimating Equation and Self-Normalization

We substitute the historical-assisted outcome regressions into the Kallus–Mao estimating structure and normalize each weighted adjustment over the full target sample.

Let $k(i)$ denote the evaluation fold containing observation i. Write its out-of-fold propensity predictions and projected outcome prediction as

$$\begin{aligned} \widehat{e}_i &= \widehat{e}_{k(i)}(X_i), \\ \widehat{r}_i &= \widehat{r}_{k(i)}(T_i, X_i, S_i), \\ \widehat{\mu}_{t,i} &= \widehat{\mu}^E_{t,k(i)}(X_i). \end{aligned}$$

Probability predictions include the clipping used in the numerical implementation. Expanding $\widehat{m}^{E}_{t,k(i)}(X_i, S_i) = \widehat{h}_H(X_i, S_i) + \widehat{\Delta}_{t,k(i)}$, the implemented estimator is

$$\begin{aligned}
\widehat{\tau}_{\mathrm{HARE-C}} = \frac{1}{N_E}\sum_{i=1}^{N_E}\left(\widehat{\mu}_{1,i} - \widehat{\mu}_{0,i}\right) \\
+ \frac{\sum_i \frac{T_i}{\widehat{e}_i}\left\{\widehat{h}_H(X_i, S_i) + \widehat{\Delta}_{1,k(i)} - \widehat{\mu}_{1,i}\right\}}{\sum_i \frac{T_i}{\widehat{e}_i}} \\
- \frac{\sum_i \frac{1-T_i}{1-\widehat{e}_i}\left\{\widehat{h}_H(X_i, S_i) + \widehat{\Delta}_{0,k(i)} - \widehat{\mu}_{0,i}\right\}}{\sum_i \frac{1-T_i}{1-\widehat{e}_i}} \\
+ \frac{\sum_{i:R_i=1} \frac{T_i}{\widehat{e}_i\widehat{r}_i}\left\{Y_i - \widehat{h}_H(X_i, S_i) - \widehat{\Delta}_{1,k(i)}\right\}}{\sum_{i:R_i=1} \frac{T_i}{\widehat{e}_i\widehat{r}_i}} \\
- \frac{\sum_{i:R_i=1} \frac{1-T_i}{(1-\widehat{e}_i)\widehat{r}_i}\left\{Y_i - \widehat{h}_H(X_i, S_i) - \widehat{\Delta}_{0,k(i)}\right\}}{\sum_{i:R_i=1} \frac{1-T_i}{(1-\widehat{e}_i)\widehat{r}_i}}.
\end{aligned} \tag{3.6}$$

Unrestricted sums run over all target observations, whereas the last two ratios use only observations with an observed long-term outcome. Each ratio divides a weighted sum by the corresponding total weight. All denominators are computed across the full target sample after combining the out-of-fold predictions.

The estimator retains the conditional-mean contrast, the surrogate adjustment, and the observed-outcome residual correction. Historical information enters through $\widehat{h}_H$, while target labels enter both the fitted residual constants and the final weighted outcome residuals.

3.4 Cross-Fitting and Standard Errors

The numerical study uses five-fold cross-fitting, stratified by treatment status and label observability. The historical model is trained on the independent historical sample and shared across target folds. Target probability-regression coefficients, residual corrections, and conditional projections are fitted in the corresponding training folds. Evaluation-fold predictions are combined by observation index before computing Equation (3.6).

For the standard-error calculation, expand each ratio in Equation (3.6) into observation-level summands. Let $\widehat{\Gamma}^{H}_i$ denote N_E times the combined contribution of observation i, so that $\widehat{\tau}_{\mathrm{HARE-C}} = N_E^{-1}\sum_i \widehat{\Gamma}^{H}_i$. The

numerical implementation uses the following raw-score plug-in standard error:

$$\widehat{\mathrm{se}}_{\mathrm{plug}} = \left[\frac{1}{N_E(N_E-1)} \sum_{i=1}^{N_E} \left(\widehat{\Gamma}_i^H - \widehat{\tau}_{\mathrm{HARE-C}} \right)^2 \right]^{1/2}, \tag{3.7}$$

with nominal 95% intervals

$$\widehat{\tau}_{\mathrm{HARE-C}} \pm 1.96\, \widehat{\mathrm{se}}_{\mathrm{plug}}.$$

This standard error is computed from the sample variance of individual contributions. Finite-sample interval calibration is evaluated in the simulations using empirical coverage and comparisons between estimated standard errors and empirical standard deviations.

4. Analytical Results

4.1 Correcting Historical-Index Bias with Target Outcomes

When the historical-index effect differs from the true long-term treatment effect, a historical-index estimator is biased for the latter (Athey et al., 2026, Theorem 4). We derive the discrepancy and show how observed target outcomes can remove it through a weighted residual correction.

Let $h(x, s)$ be a fixed historical prediction function, and let τ_h denote the average treatment effect on this prediction. The true long-term effect and the historical-index effect are, respectively,

$$\begin{aligned} \tau &= \mathbb{E}_E[Y(1) - Y(0)], \\ \tau_h &= \mathbb{E}_E[h(X, S(1)) - h(X, S(0))]. \end{aligned} \tag{4.1}$$

Subtracting the two expressions and regrouping by treatment state gives

$$\begin{aligned} \tau - \tau_h &= \mathbb{E}_E[Y(1) - Y(0)] - \mathbb{E}_E[h(X, S(1)) - h(X, S(0))] \\ &= \mathbb{E}_E[Y(1) - h(X, S(1))] - \mathbb{E}_E[Y(0) - h(X, S(0))]. \end{aligned} \tag{4.2}$$

Thus, the difference between the two effects is the difference between the mean historical-prediction residuals under treatment and control.

Define the target conditional-mean discrepancy relative to h as

$$d_t^h(x, s) = m_t^E(x, s) - h(x, s).$$

Under consistency and unconfounded treatment assignment, $m_t^E(x, s) = \mathbb{E}_E[Y(t) \mid X = x, S(t) = s]$. Applying iterated expectations therefore yields

$$\begin{aligned} \mathbb{E}_E[Y(t) - h(X, S(t))] &= \mathbb{E}_E[\mathbb{E}_E[Y(t) \mid X, S(t)] - h(X, S(t))] \\ &= \mathbb{E}_E[m_t^E(X, S(t)) - h(X, S(t))] \\ &= \mathbb{E}_E[d_t^h(X, S(t))]. \end{aligned} \tag{4.3}$$

Combining Equations (4.2) and (4.3),

$$\tau - \tau_h = \mathbb{E}_E[d_1^h(X, S(1))] - \mathbb{E}_E[d_0^h(X, S(0))].$$

Although target long-term outcomes are only partially observed, this residual contrast is identifiable under the missing-at-random and positivity conditions. Define the signed treatment weight

$$a(T, X) = \frac{T}{e(X)} - \frac{1 - T}{1 - e(X)}.$$

Missing at random implies

$$\begin{aligned}\mathbb{E}_E\left[\frac{R}{r(T, X, S)}\{Y - h(X, S)\}\,\middle|\, T, X, S\right] \\ = \mathbb{E}_E[Y - h(X, S) \mid T, X, S] = d_T^h(X, S).\end{aligned}$$

Applying treatment inverse-probability weighting then gives

$$\begin{aligned}&\mathbb{E}_E\left[a(T, X)\frac{R}{r(T, X, S)}\{Y - h(X, S)\}\right] \\ &= \mathbb{E}_E\left[\frac{T}{e(X)}d_1^h(X, S) - \frac{1 - T}{1 - e(X)}d_0^h(X, S)\right] \\ &= \mathbb{E}_E[d_1^h(X, S(1))] - \mathbb{E}_E[d_0^h(X, S(0))] \\ &= \tau - \tau_h.\end{aligned} \tag{4.4}$$

Observed target outcomes therefore identify exactly the mean residual contrast omitted by the historical-index effect.

To construct the corresponding estimator, consider a reference version of HARE with known treatment and labeling probabilities and unnormalized weights. Write $a_i = a(T_i, X_i)$, $r_i = r(T_i, X_i, S_i)$, and $h_i = h(X_i, S_i)$. The uncorrected historical-index estimator is

$$\widehat{\tau}_h = \frac{1}{N_E}\sum_{i=1}^{N_E} a_i h_i.$$

Adding the empirical target residual correction gives

$$\boxed{\widehat{\tau}_{\mathrm{HARE}}^{\mathrm{ref}} = \widehat{\tau}_h + \frac{1}{N_E}\sum_{i=1}^{N_E} a_i \frac{R_i}{r_i}(Y_i - h_i).} \tag{4.5}$$

The residual contribution is defined as zero when $R_i = 0$.

Proposition 1 (Unbiasedness of the reference HARE estimator)**.** *Suppose Assumptions A1–A4 hold, h is square-integrable under both target potential-surrogate distributions, and the treatment and labeling probabilities are known. The uncorrected historical-index estimator has bias*

$$\mathbb{E}_E[\widehat{\tau}_h] - \tau = \tau_h - \tau.$$

The reference HARE estimator in Equation (4.5) *removes this bias:*

$$\boxed{\begin{aligned}\mathbb{E}_E[\widehat{\tau}_{\mathrm{HARE}}^{\mathrm{ref}}] &= \tau, \\ \mathbb{E}_E[\widehat{\tau}_{\mathrm{HARE}}^{\mathrm{ref}}] - \tau &= 0.\end{aligned}} \tag{4.6}$$

If h is trained on an independent historical sample, these statements hold conditional on that sample.

Proof. Under unconfounded treatment assignment, $\mathbb{E}_E[a(T,X)h(X,S)] = \tau_h$, so $\mathbb{E}_E[\widehat{\tau}_h] = \tau_h$. Using Equation (4.4),

$$\begin{aligned}\mathbb{E}_E[\widehat{\tau}^{\text{ref}}_{\text{HARE}}] &= \mathbb{E}_E[\widehat{\tau}_h] + \mathbb{E}_E\left[\frac{1}{N_E}\sum_{i=1}^{N_E} a_i \frac{R_i}{r_i}(Y_i - h_i)\right] \\ &= \tau_h + \mathbb{E}_E\left[a(T,X)\frac{R}{r(T,X,S)}\{Y - h(X,S)\}\right] \\ &= \tau_h + (\tau - \tau_h) = \tau.\end{aligned}$$

Hence the historical-index bias is exactly canceled in expectation by the target residual correction. □

Proposition 1 establishes the bias-correction role of target outcomes. Even when the historical prediction yields an index effect different from the true long-term effect, the HARE estimator recovers the omitted effect through target residual correction. Unlike the surrogate-index approach, this correction does not require statistical surrogacy or cross-population comparability. Instead, it uses observed target long-term outcomes, together with the target identification conditions, to account for discrepancies between historical prediction and the target outcome relationship.

The HARE estimator developed in Section 3 incorporates this correction principle into the Kallus–Mao estimating structure with surrogate adjustment, estimated nuisance functions, cross-fitting, and self-normalized weights. Section 5 further evaluates whether this bias-correction mechanism translates into improved finite-sample ATE estimation performance.

4.2 How Historical Information Reduces the Target Learning Burden

Proposition 1 establishes the bias-correction role of target outcomes. Historical information can also reduce the amount of outcome structure that must be learned from target labels. Under $m_t^E = h_H + d_t$, historical-assisted learning uses the historical prediction as a baseline and fits the remaining discrepancy from observed target outcomes.

Structural Conditions H1 and H2 in Section 2.4 describe the historical accuracy and residual simplicity that motivate this approach. As shown in Equation (2.19), the residual relationship in practice is

$$m_t^E - \widehat{h}_H = d_t + (h_H - \widehat{h}_H).$$

The following benchmark isolates the learning gain by taking h_H as known and assuming that the discrepancy is exactly represented by a lower-dimensional correction space. It allows nonconstant corrections.

Fix a treatment state t and condition on the covariate and surrogate values of n_t labeled observations in that arm. Throughout this benchmark, $\mathbb{E}_{\text{tr}}$ and Var_{tr} refer to randomness in the observed outcomes, with these design points held fixed. Suppose

$$\begin{aligned} Y_i &= m_t^E(X_i, S_i) + \varepsilon_i, \qquad i = 1, \ldots, n_t, \\ \mathbb{E}_{\text{tr}}[\varepsilon_i] &= 0, \qquad \text{Var}_{\text{tr}}(\varepsilon_i) = \sigma_t^2 > 0, \end{aligned} \tag{4.7}$$

with independent errors conditional on the fixed design.

For this benchmark, specialize the classes $\mathcal{F}$ and $\mathcal{G}$ from Section 2.4 to finite-dimensional linear spaces of functions. Suppose

$$m_t^E \in \mathcal{F}, \qquad d_t \in \mathcal{G}, \qquad p = \dim(\mathcal{F}), \qquad q = \dim(\mathcal{G}), \qquad 1 \le q < p \le n_t.$$

Both spaces are prespecified independently of these target outcomes, and their design matrices have full column rank. The basis functions may include nonlinear transformations and interactions; linearity refers to the coefficients estimated from target labels.

Let $\widehat{m}_t^{\text{direct}}$ be the ordinary least-squares fit to Y using $\mathcal{F}$. Let $\widehat{d}_t$ be the ordinary least-squares fit to $Y - h_H(X, S)$ using $\mathcal{G}$, and define

$$\widehat{m}_t^{\text{HARE}} = h_H + \widehat{d}_t.$$

Both estimators are evaluated by their expected average squared errors for the true conditional means at the fixed design points, rather than by their residual sums of squares against the observed outcomes.

Proposition 2 (Target-label efficiency in a finite-dimensional benchmark)**.** *Under the conditions above,*

$$\begin{aligned} \mathbb{E}_{\text{tr}}\left[\frac{1}{n_t}\sum_{i=1}^{n_t}\left\{\widehat{m}_t^{\text{direct}}(X_i, S_i) - m_t^E(X_i, S_i)\right\}^2\right] &= \frac{\sigma_t^2 p}{n_t}, \\ \mathbb{E}_{\text{tr}}\left[\frac{1}{n_t}\sum_{i=1}^{n_t}\left\{\widehat{m}_t^{\text{HARE}}(X_i, S_i) - m_t^E(X_i, S_i)\right\}^2\right] &= \frac{\sigma_t^2 q}{n_t}. \end{aligned} \tag{4.8}$$

At a common target-label budget, the ratio of the historical-assisted error to the direct-learning error is therefore

$$\frac{\sigma_t^2 q / n_t}{\sigma_t^2 p / n_t} = \frac{q}{p} < 1. \tag{4.9}$$

Proof. Under correct specification, ordinary least squares projects the outcome noise onto the fitted-value space. Choose an orthonormal basis for that space. Along each basis direction, the noise projection has mean zero and variance σ_t^2. The direct fit has p such directions, so its expected squared conditional-mean error summed over the design points is $\sigma_t^2 p$.

Subtracting the known historical function leaves a correctly specified regression for d_t with the same outcome noise. Its fitted-value space has q directions, giving total expected squared error $\sigma_t^2 q$. Dividing each expression by n_t proves Equation (4.8); their ratio gives Equation (4.9). □

The proposition shows how historical information reduces the number of degrees of freedom that must be estimated from target labels. In this benchmark, learning a q-dimensional discrepancy rather than a p-dimensional outcome relationship yields a smaller conditional-mean estimation error at the same label budget.

To express the comparison in terms of label requirements, let $\epsilon^2 > 0$ be the desired level of average squared estimation error. Holding the model dimensions fixed, and provided full-rank designs are available at the relevant sample sizes, direct learning attains this threshold when

$$n_t \ge \max\left\{p, \left\lceil \frac{\sigma_t^2 p}{\epsilon^2} \right\rceil\right\}, \tag{4.10}$$

whereas historical-assisted learning attains it when

$$n_t \geq \max\left\{q, \left\lceil \frac{\sigma_t^2 q}{\epsilon^2} \right\rceil\right\}. \tag{4.11}$$

Here $\lceil \cdot \rceil$ denotes rounding up to the nearest integer. Before integer rounding, the two requirements are $p \max\{1, \sigma_t^2/\epsilon^2\}$ and $q \max\{1, \sigma_t^2/\epsilon^2\}$, respectively, and their ratio is q/p. Thus, in this benchmark, historical information improves outcome-regression accuracy at a fixed target-label budget and reduces the target-label requirement for the same fixed-design accuracy.

The exact comparison assumes a known historical function and zero approximation error in the correction space. In the full HARE setting, historical estimation error and imperfect approximation of d_t also contribute to outcome-learning error, as described by Structural Conditions H1 and H2.

The result concerns outcome-regression learning. In the orthogonal estimating equation of Kallus and Mao (2025, Lemma 3.1), outcome-regression errors enter the bias bound through products with propensity-estimation errors. Section 5 evaluates whether historical assistance improves finite-sample ATE RMSE and target-label efficiency in the implemented estimator.

5. Simulation Study

5.1 Design and Implementation

We conducted Monte Carlo simulations in R on an Alibaba Cloud server to evaluate whether historical outcome information improves long-term ATE estimation and reduces the target-label budget needed to attain comparable accuracy. Outcome models were fitted using the `gbm` package, stratified folds were constructed with `splitTools`, and simulation jobs were executed in parallel using `future` and `future.apply`.

Table 2 summarizes the main design. The five label budgets and three discrepancy settings yield 15 configurations, each with 50 successful repetitions per method in the reported results. Label budgets specify expected rather than fixed numbers of observed target outcomes.

Treatment probabilities follow a logistic function of the six pretreatment covariates and are clipped to $[0.05, 0.95]$. The labeling log-odds depend on $0.40X_1 - 0.40X_2 + 0.25X_3^2 - 0.25X_4^2$, with an intercept calibrated so that the mean labeling probability over the generated target covariates equals the specified budget divided by N_E. Treatment and labeling indicators are then drawn from their respective Bernoulli distributions.

The historical conditional mean $h_H(X, S)$ contains sine, quadratic, hyperbolic-tangent, and interaction terms. Historical outcomes equal this conditional mean plus standard normal noise. Target outcomes are

$$Y(t) = h_H(X, S(t)) + \Delta_t + \lambda u_t(X, S(t)) + \varepsilon_t, \tag{5.1}$$

where $\Delta_1 = 0.25$, $\Delta_0 = -0.25$, and ε_t is standard normal noise. The nonlinear functions u_t are centered under their corresponding target potential-surrogate distributions and approximately scaled to unit standard deviation using 200,000 additional simulated observations. Their functional forms are held fixed across λ settings. Table 3 reports the population quantities specified by the design.

Table 4 distinguishes the information and outcome-regression construction used by each method. The three feasible implementations are compared on the same target observations and label assignments. The oracle uses the true nuisance functions with the same observed-outcome pattern. Weighted adjustments are self-normalized over the full target sample as described in Section 3.3.

Table 2. Main simulation design and sampling scheme

Component	Specification
Historical sample	$N_H = 8{,}000$; observed variables: X, S, Y.
Target sample	$N_E = 8{,}000$; X, T, S, R observed for all units; Y observed only when $R = 1$.
Pretreatment covariates	Six independent standard normal coordinates, $X_j \sim \mathcal{N}(0, 1)$, in both populations.
Target potential surrogates	Five coordinates per treatment state: $S_j(t) \sim \mathcal{N}(2t - 1, 1)$, $j = 1, \ldots, 5$; generated independently of X and one another.
Outcome noise	Independent standard normal errors in the historical and target outcome equations.
Expected target-label budgets	100, 200, 400, 800, 1,600.
Corresponding expected label rates	1.25%, 2.5%, 5%, 10%, 20%.
Discrepancy settings	$\lambda = 0,\ 0.5,\ 1$; only the amplitude of the heterogeneous discrepancy changes across settings.
Monte Carlo repetitions	50 per configuration; 15 scenario–budget configurations.
Cross-fitting	Five folds, stratified jointly by treatment T and label observability R.
Reuse of random draws	Within a budget and repetition, the three λ settings share historical data and target random draws. Samples are regenerated across budgets.

Note. The observed target surrogate vector is $S = S(T)$. Historical surrogates use the same conditional mixture of the two surrogate distributions, generated with a latent assignment having probability $e(X)$; that assignment is not included in the historical outcome regression.

Table 3. Population quantities across discrepancy settings

Quantity	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$
Mean treated discrepancy, $\mathbb{E}_E[d_1(X, S(1))]$	0.25	0.25	0.25
Mean control discrepancy, $\mathbb{E}_E[d_0(X, S(0))]$	−0.25	−0.25	−0.25
Discrepancy SD in either arm	0	≈ 0.5	≈ 1
Historical-index effect, τ_h	1.5	1.5	1.5
Target long-term ATE, τ	2	2	2
Effect omitted by the index, $\tau - \tau_h$	0.5	0.5	0.5
Index estimand bias, $\tau_h - \tau$	−0.5	−0.5	−0.5

Note. Here τ_h is the effect on the true historical conditional mean h_H. Entries are population design values, not Monte Carlo estimates. Discrepancy SD is evaluated under $(X, S(t))$ for each t and is approximate because the scaling constants are estimated from the calibration sample. Increasing λ leaves the mean effect gap unchanged.

Table 5 reports the learning settings. Historical outcome GBMs use all 3,000 trees, whereas target outcome and conditional-projection GBMs select the number of trees by out-of-bag (OOB) performance. The probability regressions and clipping thresholds are held fixed across the feasible methods that use them.

The primary performance measures are empirical bias and root mean squared error relative to the population truth $\tau = 2$. Let $B = 50$ be the number of repetitions and $\widehat{\tau}_b$ the estimate in repetition b. We compute

$$\widehat{\text{Bias}} = \frac{1}{B}\sum_{b=1}^{B}(\widehat{\tau}_b - \tau),$$
$$\widehat{\text{RMSE}} = \left[\frac{1}{B}\sum_{b=1}^{B}(\widehat{\tau}_b - \tau)^2\right]^{1/2}. \tag{5.2}$$

Table 4. Compared estimators and outcome-learning implementations

Method	Information used	Outcome learning and estimation
SI-AIPW-GBM	Historical X, S, Y; all target X, T, S.	Fit the historical outcome model and project its predictions on X within each target treatment arm. Estimate the augmented historical-index effect without a target-outcome residual correction.
SA-EIF-GBM	Labeled and unlabeled target observations.	Fit m_t^E on (X, S) and μ_t^E on X separately using labeled observations in each arm. Use the surrogate-assisted estimator with target-outcome residual correction.
HARE-C	Historical sample and both target subsamples.	Fit the historical predictor, estimate an unweighted residual mean within each labeled training arm, and project historical predictions on X using all target training observations in that arm. Use the same target-outcome estimating structure.
SA-EIF-Oracle	Target observations and the true nuisance functions.	Supply the true e, r, m_t^E, and μ_t^E to the same self-normalized estimator; retain the original target label assignments.

Note. The oracle is an infeasible reference. The feasible methods share the relevant fitted probability models; SI-AIPW-GBM does not use the labeling model. All fitted target functions are evaluated out of fold. Predictor expansions and tuning rules are reported in Table 5.

Table 5. Learning algorithms, predictor sets, and tuning rules

Setting	Specification
Panel A. Outcome and projection GBMs	
Implementation and loss	`gbm::gbm`; Gaussian squared-error loss.
Outcome-model predictors	$X_1, \ldots, X_6$, their squares, and $S_1, \ldots, S_5$ for historical and target outcome models.
Baseline/projection predictors	$X_1, \ldots, X_6$ and their squares.
Maximum number of trees	3,000.
Historical tree selection	Fixed at 3,000 trees.
Target/projection tree selection	OOB selection, restricted to 50–3,000 trees; 3,000 if the OOB choice is unavailable or nonfinite.
Interaction depth	3.
Learning rate	0.03.
Subsampling fraction	0.8.
Minimum terminal-node size	5 observations.
Panel B. Probability models and residual correction	
Treatment probability	Logistic regression on $X_1, \ldots, X_6$, without an intercept; predictions clipped to $[0.05, 0.95]$.
Labeling probability	Logistic regression on $X_1, \ldots, X_6$ and their squares, with an intercept and offset $\text{logit}(N_L/N_E)$; predictions clipped to $[10^{-6}, 1]$.
HARE-C residual model	Unweighted intercept-only least squares within each labeled treatment arm of the training fold.

Note. GBM settings apply to the supplied training fold, except for the historical model, which is trained on the independent historical sample. The labeling offset uses the overall observed target-label fraction; regression coefficients are estimated within each training fold.

We also record empirical standard deviations, estimated standard errors, interval coverage, and interval widths. Bias and RMSE use the population ATE, rather than the realized sample average of $Y_i(1) - Y_i(0)$.

5.2 Bias Correction Relative to the Surrogate-Index Benchmark

We next focus on the main comparison for the constant-discrepancy configuration ($\lambda = 0$): whether historical-assisted residual estimation corrects the bias of the surrogate-index benchmark. Table 6 reports the bias of the surrogate-index estimator and HARE-C across five target-label budgets. The pattern is clear and highly stable. The surrogate-index benchmark remains downward biased by about -0.52 at every budget, whereas the bias of HARE-C stays close to zero throughout. Measured in absolute value, the bias reduction delivered by HARE-C ranges from 89.7% to 98.3% across the five budgets.

At a target-label budget of 800, for example, the bias of the surrogate-index benchmark is -0.5201, while the bias of HARE-C is 0.0103. This comparison shows that the main gain of HARE-C in this design is the correction of the historical-index bias. Figure 1 visualizes this comparison for the 800-label case by plotting the Monte Carlo mean estimates of the two methods against the true ATE. Error bars report the empirical standard deviation across replications.

Table 6. Bias comparison under constant discrepancies ($\lambda = 0$)

Target-label budget	SI-AIPW-GBM bias	HARE-C bias	Absolute-bias reduction (%)
100	-0.5331	0.0092	98.28
200	-0.5299	-0.0548	89.66
400	-0.5230	-0.0293	94.40
800	-0.5201	0.0103	98.01
1600	-0.5280	0.0093	98.24

Notes. The surrogate-index benchmark is recorded as `OI-GBM` in the simulation output table. The final column reports $100 \times \{|\text{Bias}_{\text{SI}}| - |\text{Bias}_{\text{HARE}}|\}/|\text{Bias}_{\text{SI}}|$. All numbers are based on the main simulation in Section 5.1 with 50 Monte Carlo replications per configuration.

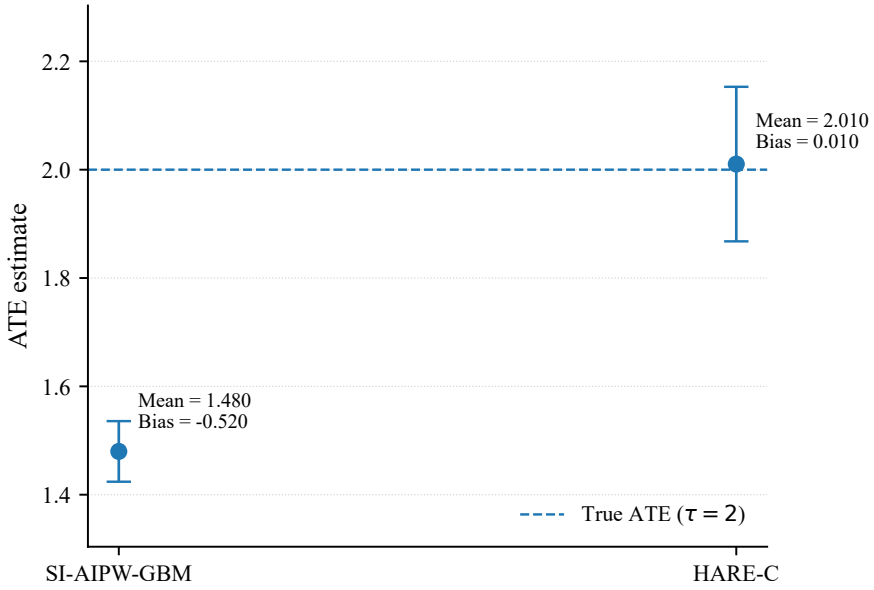


Figure 1. Monte Carlo mean ATE estimates at target-label budget 800 under constant discrepancies ($\lambda = 0$)

Note. Points report the Monte Carlo mean estimate over 50 replications. Error bars show the empirical standard deviation of the estimator across replications, not confidence intervals. The dashed horizontal line denotes the true population ATE, $\tau = 2$.

5.3 RMSE and Target-Label Efficiency Relative to Target-Only Estimation

We next compare HARE-C with the target-only estimator, SA-EIF-GBM, in terms of ATE RMSE and target-label efficiency. The comparison uses the constant-discrepancy configuration ($\lambda = 0$), with $N_H = N_E = 8{,}000$ and the same target observations and label assignments for both methods within each repetition.

Table 7 reports the empirical RMSE at each expected target-label budget. HARE-C attains lower RMSE than SA-EIF-GBM at all five budgets. At a budget of 800, their RMSEs are 0.1416 and 0.2392, respectively, corresponding to a reduction of 40.8%. Across the five budgets, the relative reduction ranges from 24.2% to 58.7%.

Table 7. ATE RMSE relative to target-only estimation under constant discrepancies ($\lambda = 0$)

Expected target-label budget	SA-EIF-GBM RMSE	HARE-C RMSE	RMSE reduction (%)
100	0.8691	0.3593	58.66
200	0.7658	0.5270	31.19
400	0.4014	0.2867	28.59
800	0.2392	0.1416	40.81
1,600	0.1549	0.1175	24.16

Note. RMSE is calculated relative to the population ATE, $\tau = 2$, over 50 Monte Carlo repetitions per configuration. The final column reports $100 \times (1 - \mathrm{RMSE}_{\mathrm{HARE-C}}/\mathrm{RMSE}_{\mathrm{SA-EIF-GBM}})$, calculated using unrounded values. Budgets refer to expected target-label counts. HARE-C additionally uses the historical sample of 8,000 observations.

The comparison across budgets also illustrates the value of historical information for reducing target-label requirements. HARE-C has an RMSE of 0.1416 at an expected budget of 800, whereas SA-EIF-GBM has an

RMSE of 0.1549 at a budget of 1,600. Mean realized label counts are 806.6 and 1,598.54, respectively. Thus, with historical data available, HARE-C attains an empirical RMSE no greater than that of the target-only comparator using half the expected target-label budget. This comparison concerns the acquisition of target long-term outcomes, rather than the total number of outcome observations across both samples.

Figure 2 displays the RMSE learning curves. Between budgets of 400 and 1,600, HARE-C's RMSE falls from 0.2867 to 0.1175, while SA-EIF-GBM's falls from 0.4014 to 0.1549. The relative RMSE reduction narrows from 40.8% at 800 labels to 24.2% at 1,600 labels, as the target-only estimator improves with additional target outcomes.

At the lowest budgets, the empirical HARE-C curve is not monotone: its RMSE is 0.3593 at a budget of 100 and 0.5270 at 200, before falling to 0.2867 at 400. The learning curves retain these observed values. Together, the comparisons show both improved accuracy at a common target-label budget and an empirical label-efficiency gain under the constant-discrepancy design.

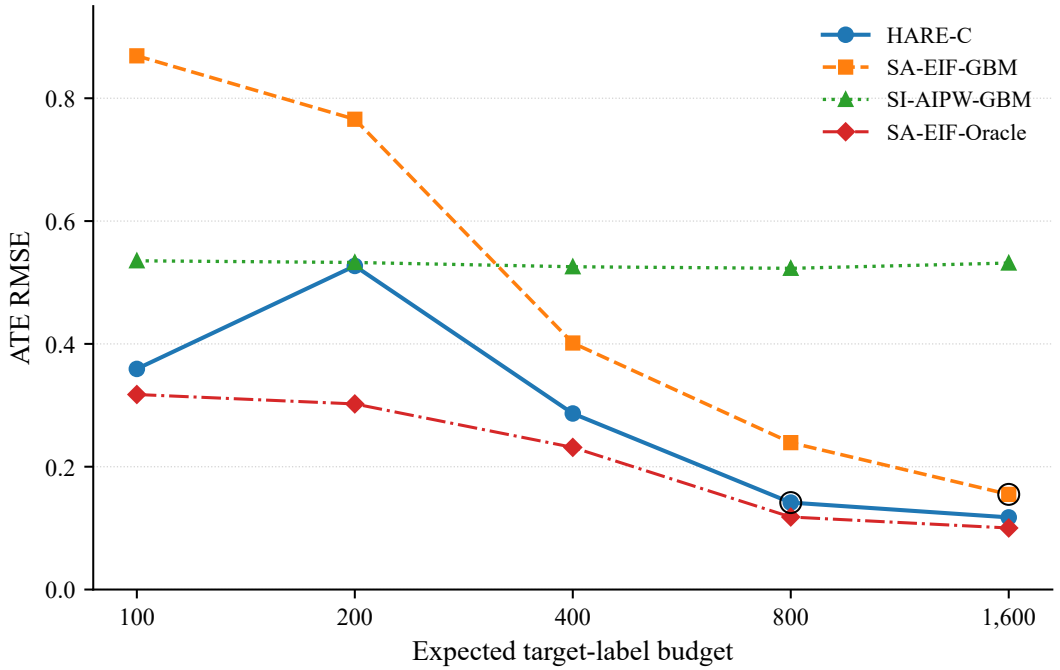


Figure 2. ATE RMSE across target-label budgets under constant discrepancies ($\lambda = 0$)

Note. The configuration is $\lambda = 0$, with $N_H = N_E = 8{,}000$ and $\tau = 2$. Each point reports RMSE over 50 Monte Carlo replications. The horizontal axis shows expected target-label budgets; realized label counts are random. Open circles highlight HARE-C at a budget of 800 and SA-EIF-GBM at a budget of 1,600, the cross-budget comparison discussed in the text. Samples are regenerated across budgets and are not nested.

5.4 Robustness to Heterogeneous Target Discrepancies

Next, we hold HARE-C's constant correction model fixed while increasing heterogeneity in the true discrepancy from $\lambda = 0$ to 0.5 and 1. This design examines the ability of a fixed correction model to accommodate heterogeneous transport. Historical prediction still contains the shared nonlinear structure, but target discrepancies that cannot be captured by a constant become larger. The true ATE and historical-index effect remain fixed across settings, separating the mean effect gap from correction-model misspecification.

Table 8 summarizes the effect of increasing discrepancy heterogeneity at an expected target-label budget of 800. The table separates HARE-C's empirical bias from its sampling variability and compares its RMSE with that of SA-EIF-GBM.

Table 8. Estimation performance under heterogeneous discrepancies at an expected target-label budget of 800

λ	HARE-C			SA-EIF-GBM RMSE	Relative RMSE reduction (%)
	Bias	Empirical SD	RMSE		
0	0.0103	0.1426	0.1416	0.2392	40.8
0.5	0.0153	0.1730	0.1719	0.2612	34.2
1	0.0203	0.2255	0.2241	0.2905	22.8

Note. Each configuration uses $N_H = N_E = 8{,}000$ and 50 Monte Carlo repetitions. The true ATE is $\tau = 2$ and the historical-index effect is $\tau_h = 1.5$ in all three settings. The final column reports $100 \times (1 - \mathrm{RMSE}_{\mathrm{HARE-C}}/\mathrm{RMSE}_{\mathrm{SA-EIF-GBM}})$, calculated from unrounded RMSE values. The HARE-C correction model is held fixed across settings, while λ increases the magnitude of the heterogeneous component of the target discrepancy.

As shown in Table 8, HARE-C's relative RMSE reduction declines from 40.8% at $\lambda = 0$ to 34.2% at $\lambda = 0.5$ and 22.8% at $\lambda = 1$. At the same time, HARE-C's empirical bias remains close to zero, while its empirical standard deviation increases from 0.1426 to 0.2255. Thus, at this label budget, the deterioration in HARE-C's performance as the discrepancy becomes more heterogeneous is driven primarily by increased estimation variability rather than by the re-emergence of a large systematic bias.

Heterogeneity also affects cross-budget label-efficiency comparisons. At $\lambda = 0.5$, HARE-C's RMSE of 0.1719 at 800 labels is close to, but slightly above, SA-EIF-GBM's 0.1688 at 1,600 labels. At $\lambda = 1$, the corresponding values are 0.2241 and 0.1854. Thus, the constant-discrepancy finding that half the target-label budget attains an RMSE no greater than that of the comparator does not persist under stronger heterogeneity. Nevertheless, HARE-C has lower empirical RMSE than SA-EIF-GBM at the same budget in all 15 configurations formed by the three discrepancy settings and five budgets. Their ordering does not reverse within the tested range.

Figure 3 compares the relative RMSE of HARE-C and SA-EIF-GBM across all five target-label budgets. A ratio below one indicates lower empirical RMSE for HARE-C. The figure retains the observed differences across budgets rather than imposing a common monotone response to increasing heterogeneity.

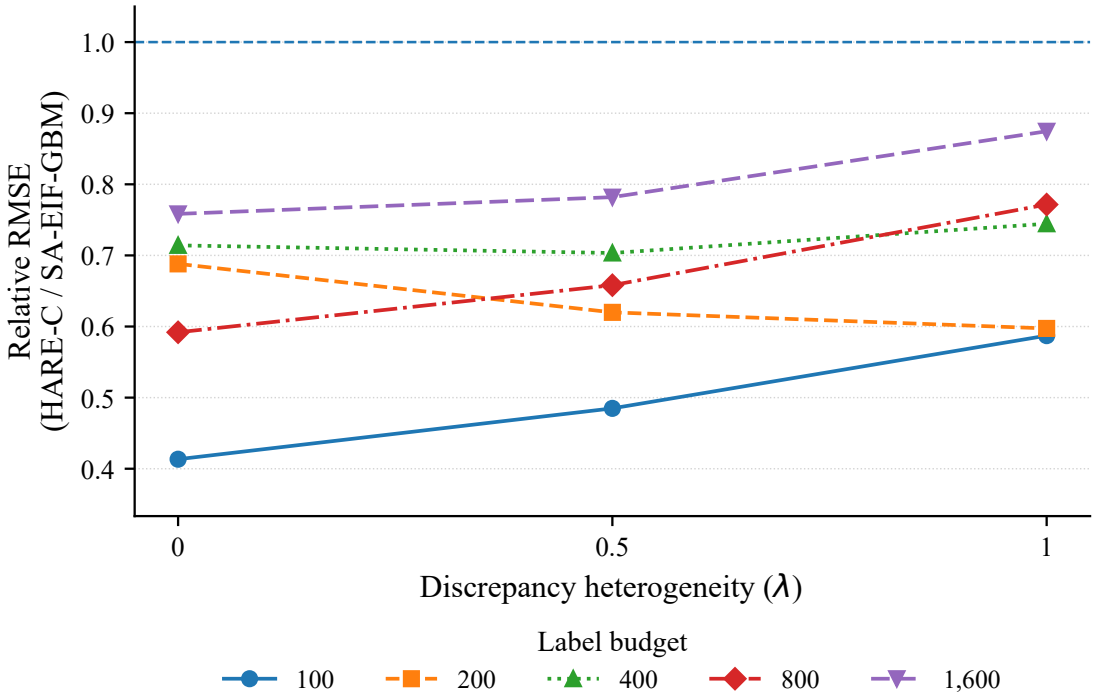


Figure 3. Relative RMSE of HARE-C to SA-EIF-GBM as target discrepancies become more heterogeneous

Note. The vertical axis reports $\mathrm{RMSE}_{\mathrm{HARE-C}}/\mathrm{RMSE}_{\mathrm{SA-EIF-GBM}}$. Each curve corresponds to one expected target-label budget. Ratios below one indicate lower empirical RMSE for HARE-C; the dashed reference line at one denotes equal RMSE. Each RMSE is computed over 50 Monte Carlo replications. The population ATE remains 2 and the historical-index effect remains 1.5 across all settings.

The variance cost of bias correction is most evident in the configuration with the fewest labels and strongest heterogeneity. At $\lambda = 1$ and a label budget of 100, HARE-C reduces empirical bias from the surrogate-index benchmark's -0.5331 to 0.0372, but its RMSE is 0.6709, above the benchmark's 0.5353. Historical-bias correction and mean squared error reduction therefore do not always move together. When target labels are very limited and discrepancies that the constant model cannot represent are substantial, the additional estimation variability can outweigh the benefit of reduced bias.

Overall, the simulations demonstrate the complementary roles of historical information and target labels. When historical data can learn the shared predictive relationship and the target discrepancy is simple, HARE-C reduces historical-index bias and the target-label requirement. As target labels become more numerous, or heterogeneity beyond the constant model becomes stronger, its relative accuracy advantage narrows in the corresponding comparisons. The gains depend on how much outcome prediction can be supplied by the historical sample and on the size of the remaining discrepancy that limited target labels must address.

6. Conclusion

This paper studies how historical outcome data and partially observed target outcomes can jointly support long-term average treatment effect estimation. HARE assigns complementary tasks to the two sources: historical data provide a baseline outcome predictor, while target labels support treatment-specific residual learning and the final weighted outcome correction. Integrating these components into the Kallus–Mao

estimating framework allows historical information to assist estimation without requiring the historical and target outcome relationships to coincide.

The analysis establishes two complementary results. Under the target identification conditions, the difference between the historical-index effect and the true long-term effect is identified by a weighted contrast of target residuals. This produces an unbiased reference estimator when treatment and labeling probabilities are known, without imposing statistical surrogacy or cross-population comparability. A separate finite-dimensional least-squares benchmark shows that, with a known historical function and a correctly specified lower-dimensional residual model, historical assistance reduces outcome-regression error and the target-label requirement for a given fixed-design accuracy.

The simulations demonstrate these complementary benefits for HARE-C. Under constant discrepancies and an expected target-label budget of 800, its empirical bias is 0.0103, compared with -0.5201 for SI-AIPW-GBM. Its RMSE is 40.8% lower than that of SA-EIF-GBM at the same budget. With 8,000 historical observations available, HARE-C at a budget of 800 also achieves lower empirical RMSE than SA-EIF-GBM at 1,600. These findings show how existing historical information can reduce the demand for additional target outcome measurements in the studied setting.

The gains depend on historical prediction quality and the remaining discrepancy structure. At the 800-label budget, stronger unmodeled heterogeneity narrows the relative RMSE reduction from 40.8% to 22.8%, primarily through increased sampling variability. With only 100 expected labels and the strongest heterogeneity, HARE-C reduces bias but has higher RMSE than the surrogate-index benchmark. The empirical evidence is limited to synthetic designs and a constant-correction implementation, while the theoretical label-efficiency comparison concerns outcome learning with known historical information. Identification continues to require unconfoundedness, missing at random, and positivity.

Future work could examine flexible residual models and applications with observed historical and target outcomes. The central implication is that historical data and target labels are complements rather than substitutes: historical information can supply reusable predictive structure, while target outcomes remain essential for learning and correcting departures from that structure.

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